

# Week 2, Lecture 1: Working with Functions

Instructor notes – Part 1: prep reading, Part 2: podium cheat sheet

This single file has two parts, both instructor-only (not a student handout). **Part 1** is read beforehand – full worked examples and the reasoning behind the flow. **Part 2** (after the page break, large print) is the podium copy.

Source material: Bolker Ch. 3, sections 3.2-3.4 (“Deterministic functions for ecological modeling”) almost verbatim in places – this lecture leans harder on the assigned reading than Week 1’s did, because today’s content is a hard prerequisite for a specific Lab 2 question (see below). Also mined: this morning’s WOD (`wod/WOD_Week2_Lecture1_RSyntaxAndNotation.Rmd`), Lab 1 Act 2 (Ricker derivative), and Lab 2 Acts 1-2 (grid search, flexible Ricker, Holling IV).

## Where this sits in the week, and a hard dependency

- Monday (this lecture): properties of deterministic functions – limits, derivatives, critical points – as tools, not yet the bestiary itself (that’s Wednesday).
- Monday AM (WOD, already done): bug-hunts in R function syntax (argument order/matching) plus deriving and coding a seasonal sine function – both are direct rehearsals for today.
- Tuesday (Lab 2): Act 1 is a grid-search optimizer for the Ricker function (algorithms, not functions per se – a callback to Week 1). **Act 2 explicitly instructs students to use “the same ‘throw out the lower-order terms’ trick from Monday’s lecture”** to find the limiting behavior of the Holling type IV function as  $x \rightarrow \infty$ . That phrase is not a coincidence – it’s this lecture. If Section 2 below (limits) doesn’t land clearly today, that Lab 2 question will stall.
- Wednesday: Bestiary of functions (Bolker 3.5) – the specific named catalog (power, exponential, hyperbolic, sigmoid...). Today builds the toolkit; Wednesday populates it.

## Scientific workflow: the modeling loop (carried over from Week 1 Lecture 2)

This was meant to open Week 1 Lecture 2 but got skipped there for time – it belongs here instead, since it’s foundational framing rather than content specific to last lecture’s algorithms-vs-closed-form material.

Worth being precise about where this comes from, since it’s easy to misremember as more

textbook-sourced than it is. Bolker section 1.6, “Outline of the modeling process” (Figure 1.5, printed p. 27 – PDF page 35, remember the +8 front-matter offset), lays out an explicit flow: *ask ecological questions* → *collect data* → *{stochastic model(s), deterministic model(s)}* → *estimate parameters* → *estimate confidence regions* → *hypothesis testing/selection* → *answer questions*, looping back to the question step (with a second dotted arrow: model selection can also loop back to the data-collection step).

Two things about that figure shaped how it’s adapted for the whiteboard:

- It genuinely starts with the *question*, not raw data. Bolker’s own footnote on p. 26 says it outright: “In an ideal world, you would identify ecological questions before you designed your experiments and gathered data (!), but in this book I will assume you’ve already got data . . . to work with.” Data comes second in his figure only because the book assumes a dataset already in hand, not because data should conceptually precede the question.
- Given that, and given a preference for a Popperian/hypothetico- deductive framing over a data-first or strictly Bayesian one, the whiteboard loop is:

THEORY/QUESTION → CANDIDATE MODEL(S) → DATA (fit)  
→ COMPARE → repeat

where “repeat” loops back to *revising or rejecting the theory*, not just re-tuning parameters. This is the same shape as Bolker’s own loop closure (his step vi, “put results together / return to step #1”, returns to step i, “ask ecological questions”) – just read through a falsificationist lens instead of an inductive one, and arguably truer to his own stated ideal (question before data) than a literal reading of his figure would be.

**What “COMPARE” means, concretely:** this step does double duty –

1. *Goodness of fit*: does the model’s prediction even resemble the data? (Lab 1’s SSR calculation for the Ricker curve.)
2. *Comparison against rivals, including the null*: does this candidate earn its keep relative to a simpler competitor? (Lab 1 Act 3’s Ricker-vs-null SSR comparison.)

In Popperian terms, this is the severe test: a model that survives comparison against a null and against competing candidates is *corroborated*, not proven; one that doesn’t survive is *falsified* and should be revised or discarded. This foreshadows the frequentist/ likelihood hypothesis-testing and model-selection machinery (LRT, AIC) in Weeks 6-8, and previews Bolker’s frequentist/likelihood/Bayesian tour in section 1.4 (pp. 13-22) without going into it today.

On the whiteboard: sketch the loop, then move straight into today’s topic (reading a function itself) – the analytical-vs-computational dichotomy from last lecture doesn’t need repeating here.

## Today's throughline (bridge from Week 1)

One-sentence bridge: last lecture was about *how* we get from a candidate model to an answer (algorithms vs. closed form); today is about *reading* a candidate function itself – what it does at the extremes, where it peaks, what its parameters mean – before you ever fit it to data.

Recall from Lab 1 Act 1: students already have an informal shape vocabulary (monotonic increasing/decreasing, saturating, hump-shaped, roughly flat) from eyeballing the tadpole data. Today's tools let you *prove* a shape claim algebraically instead of just eyeballing a curve – the same distinction as Bolker 3.2/3.3 vs. 3.4: numerically (plug in values, plot it, `curve()`) gives you a picture; analytically (algebra and calculus) tells you *why* the picture looks that way, and lets you reason about regions you didn't plot.

## Limits: throw out the lower-order terms (the required section)

This is Bolker's own phrasing (3.4.1), worth keeping close to verbatim since Lab 2 Act 2 quotes it back: *"The basic principle is to throw out lower-order terms. As  $x$  grows, it will eventually grow much larger than the largest constant term in the equation. Terms with larger powers of  $x$  will dwarf smaller powers, and exponentials will dwarf any power. If  $x$  is very small then you apply the same logic in reverse; constants are bigger than (positive) powers of  $x$ , and negative powers... are bigger than any constants."*

In short, a hierarchy of "who wins" as  $x \rightarrow \infty$ :

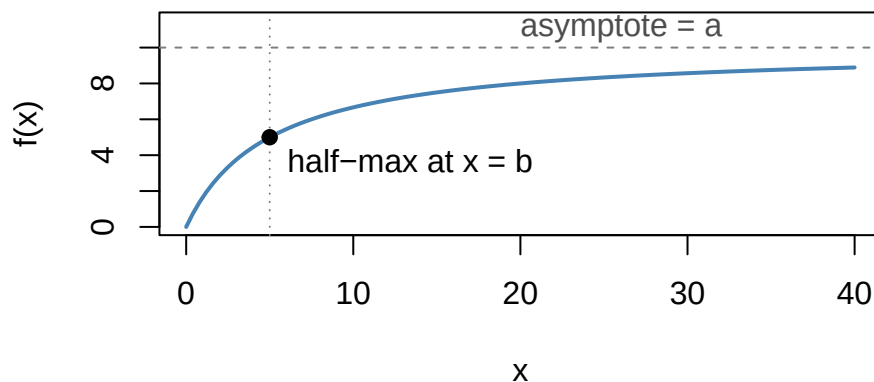
$$\text{exponential} \gg x^{\text{large power}} \gg x^{\text{small power}} \gg \text{constant}$$

and the *same* hierarchy reverses as  $x \rightarrow 0$ : constants beat positive powers of  $x$ , and negative powers of  $x$  (i.e.  $1/x$ ,  $1/x^2$ , ...) beat everything.

**Worked example (Bolker's own – Michaelis-Menten), do this on the board:**

$$f(x) = \frac{ax}{b+x}$$

- As  $x \rightarrow \infty$ :  $x \gg b$ , so  $b+x \approx x$ , so  $f(x) \approx ax/x = a$  – the curve levels off at the constant  $a$  (a horizontal asymptote).
- As  $x \rightarrow 0$ :  $b \gg x$ , so  $b+x \approx b$ , so  $f(x) \approx ax/b$  – a straight line through the origin with slope  $a/b$ .



**Sticking point to flag:** students will want to “plug in infinity.” Head that off explicitly – this is qualitative dominance reasoning (which term is negligible), not literal substitution. Also flag that the *same* parameter can play opposite roles at the two ends:  $b$  is negligible as  $x \rightarrow \infty$  but dominant as  $x \rightarrow 0$ .

**Fallback tool (mention, don’t dwell):** for fraction limits where both numerator and denominator vanish or blow up together (an indeterminate  $0/0$  or  $\infty/\infty$  form, so the dominance trick doesn’t immediately apply), L’Hopital’s rule says  $\lim \frac{a(x)}{b(x)} = \lim \frac{a'(x)}{b'(x)}$  – take the derivative of the top and bottom separately, not the derivative of the whole fraction. One line, not a worked example today.

**Do not solve Holling IV’s limit here.**  $y = ax^2/(b + cx + x^2)$  is literally Lab 2 Act 2’s question 1c – comparing the highest powers of  $x$  in numerator ( $x^2$ ) and denominator ( $x^2$ ) is the same dominance trick, one level harder (both leading terms are degree 2, so the constant  $a$  from the ratio of leading coefficients survives). Let students discover that tomorrow; today’s example (Michaelis-Menten) is deliberately a different, simpler rational function so the technique transfers without the answer being pre-given.

## Derivatives and critical points (recap + formalize)

Students already computed one derivative and one critical point by hand in Lab 1 Act 2 – this section names what they did, generalizes it, and contrasts it with a monotonic function that has no peak at all.

**Recap, don’t re-derive from scratch:** the Ricker function  $y = axe^{-bx}$  (Lab 1’s running example:  $a = 0.9$ ,  $b = 1/12$ ). Product rule:

$$\frac{dy}{dx} = ae^{-bx} + ax(-be^{-bx}) = a(1 - bx)e^{-bx}$$

Setting this to zero: since  $ae^{-bx} \neq 0$ , the only way the product is zero is  $1 - bx = 0$ , i.e.  $x^* = 1/b$ .

```
a_r <- 0.9; b_r <- 1/12
x_star <- 1 / b_r
peak_height <- a_r * x_star * exp(-b_r * x_star)
x_star
```

```
## [1] 12
```

```
peak_height
```

```
## [1] 3.973098
```

That's  $x^* = 12$  mm and peak height 3.97 – matching Lab 1 Act 3's table exactly (TBL = 12, Ricker predicted = 3.97). This is the payoff of the “critical point” concept: it's not new math, it's a name for exactly the calculation they already did.

**Formalize:** a critical point is any  $x$  where  $f'(x) = 0$  (or is undefined) – a candidate for a peak, a trough, or (at a saddle point) a flat spot that isn't either. To classify which, plug the critical  $x$ -value back into  $f$  itself (peak/trough height) and reason about the sign of  $f'$  on either side (or use the second derivative).

**Contrast with a monotonic function:** try the same exercise on Michaelis-Menten,  $f(x) = ax/(b + x)$ . Its derivative (quotient rule) is  $f'(x) = ab/(b + x)^2$  – always positive (a sum of squares in the denominator can't be negative), so the function is *monotonic increasing* everywhere and *never* has a critical point. That's exactly why it saturates toward an asymptote instead of peaking and coming back down – contrast this shape against Ricker's hump directly, on the board, side by side.

**Sticking point to flag:** “critical point” (where  $f'(x) = 0$ ) is easy to confuse with “root” (where  $f(x) = 0$ ). The Ricker function has both: it's zero at  $x = 0$  (a root) and peaks at  $x = 1/b$  (a critical point) – two different questions with two different answers from the same function.

## R convention: x first, then parameters (with defaults)

Directly from Bolker's R supplement (3.3.1, p. 104): “When you write such an R function, you would typically make the input value ( $x$ ) be the first argument, followed by all of the other parameters. It often saves time if you assign default values to the other parameters.”

This is exactly why `ricker <- function(x, a, b)` is the convention used everywhere in the labs (and why WOD 1a's `function(a, b, x)` was buggy this morning – it broke that convention on purpose). Point this out explicitly: the WOD wasn't testing an arbitrary rule, it was testing *this* rule, which is about to matter again.

**Why it matters going forward, concretely:** `apply(xvec, ricker)` only works cleanly if `x` is the first positional argument – `apply` always feeds the vector element into the first open slot. Also preview: Lab 2 Act 2 asks students to write `flex_ricker` and `holling4`

“following the same pattern as the `ricker` function (`x` first, then parameters)” – today is where that instruction gets its justification, not just its statement.

**Scale parameters, briefly (Bolker 3.4.2):** when a parameter appears as  $bx$  or  $x/c$ , its job is to stretch or shrink the curve along the  $x$ -axis without changing its shape – Bolker calls this a *characteristic scale*.  $b$  has units of inverse- $x$ ;  $c = 1/b$  has the same units as  $x$  and is often easier to interpret (e.g. “the peak is at  $x = 12$ ” instead of “ $b = 1/12$ ”). This is exactly what Lab 2 Act 2’s flexible Ricker does: it reparameterizes  $b$  as  $p$  (the peak location directly) precisely so the parameter is easier to interpret and to set. One sentence today is enough – Act 2 does the work of actually reparameterizing.

## Take-home message

- Two ways to know a function: numerically (plot it, plug in numbers) and analytically (limits, derivatives) – today was almost entirely the analytical toolkit.
- Limits at the extremes: throw out lower-order terms. Same hierarchy, reversed at the two ends (exponential  $\gg$  big power  $\gg$  small power  $\gg$  constant  $\gg$  negative power, as  $x \rightarrow \infty$ ; flip it for  $x \rightarrow 0$ ).
- Critical points ( $f'(x) = 0$ ) find peaks/troughs; monotonic functions (no sign change in  $f'$ ) never have any.
- R convention (`x` first, params after, with defaults) isn’t arbitrary style – it’s what makes `sapply()` and vectorized code work cleanly, and it’s the same convention Lab 2 asks you to follow when writing new bestiary members from scratch.

## Roadmap tie-in

- Wednesday: the bestiary itself (Bolker 3.5) – linear, power, exponential, hyperbolic/Michaelis-Menten, sigmoid families, each characterized using exactly today’s tools (limits, critical points, scale parameters).
- Tuesday’s Lab 2 Act 2 needs today’s limit technique directly (Holling IV, question 1c) – flagged above, worth a one-line reminder at the end of lecture.
- Later in the semester: “throw out lower-order terms” reasoning resurfaces informally whenever we ask what a fitted model predicts far outside the range of the data (extrapolation risk) – worth a passing mention now, not a digression.

## Discussion seed (if time remains)

Ask: *pick any function from your own field that saturates (levels off) as some predictor gets large. What’s the mechanism that caps it – what in the real world plays the role of “a” (the asymptote) in Michaelis-Menten?* Good prompt for surfacing intuition about *why* saturation happens biologically, not just how to compute it. (If it comes up: “ $b$ ” is a different question – not what caps the curve, but how fast it gets there.)

# PART 2 – PODIUM CHEAT SHEET

\*Podium copy – terse on purpose. Full derivations, rationale, and code: see Part 1 above.\*

## 0. Open (~2 min)

- Last time: *how* we get an answer (algorithm vs. closed form)
- Today: *reading* a function itself – extremes, peaks, parameters
- Recall Lab 1’s shape vocabulary (monotonic / saturating / hump) – today we prove those claims, not just eyeball them

## 1. Scientific workflow: the modeling loop (~5 min)

- **Loop (Popperian, not data-first):** THEORY/QUESTION → CANDIDATE MODEL(S) → DATA + FIT → COMPARE → repeat
- repeat = revise/reject the **theory**, not just re-tune parameters
- COMPARE = (1) fit ok? *and* (2) beats rivals/null?
- → survive = corroborated; fail = falsified
- cf. Bolker Fig. 1.5 (p.27) – his loop too, question-first

## 2. Two ways to know a function (~2 min)

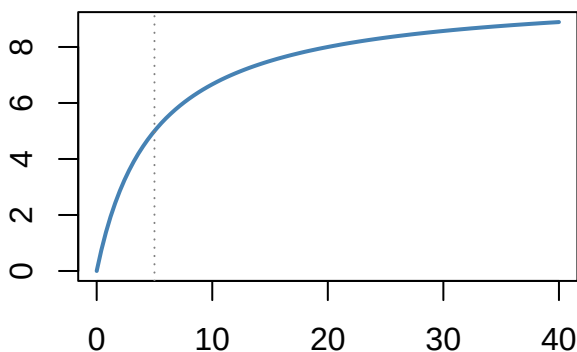
- **Numerically:** plug in values, `curve()`, plot it – a picture
- **Analytically:** limits + derivatives – *why* the picture looks that way
- Today = almost entirely analytical

### 3. Limits: throw out the lower-order terms (~10-12 min)

The rule (Bolker's own words – write it up):

- as  $x \rightarrow \infty$ : exponential  $\gg$  big power  $\gg$  small power  $\gg$  constant
- as  $x \rightarrow 0$ : reverse it – constant  $\gg$  positive power  $\gg$  nothing beats a negative power ( $1/x$ ,  $1/x^2$ , ...)

Board example – Michaelis-Menten:  $f(x) = \frac{ax}{b+x}$



- $x \rightarrow \infty$ :  $b+x \approx x \Rightarrow f(x) \approx ax/x = a$  (asymptote)
- $x \rightarrow 0$ :  $b+x \approx b \Rightarrow f(x) \approx ax/b$  (line, slope  $a/b$ )
- **Not** “plug in infinity” – dominance reasoning, term by term
- $b$  is negligible at one end, dominant at the other – same parameter, opposite roles
- Fallback only, don't dwell: L'Hopital's rule for  $0/0$  or  $\infty/\infty$  forms – derivative of top over derivative of bottom
- **Do not solve Holling IV today** – that's Lab 2 Act 2, question 1c, tomorrow. Michaelis-Menten teaches the same trick without giving away the answer.

#### 4. Derivatives and critical points (~7-8 min)

**Recap (already done in Lab 1 Act 2), don't re-derive:** Ricker  $y = axe^{-bx}$ , product rule  $\rightarrow dy/dx = a(1 - bx)e^{-bx}$

- critical point:  $1 - bx = 0 \Rightarrow x^* = 1/b$
- with  $a = 0.9$ ,  $b = 1/12$ :  $x^* = 12$ , height = 3.97 – matches Lab 1 Act 3's table exactly
- **critical point:** name for exactly what they already computed

**Contrast – Michaelis-Menten's derivative** (quotient rule):  $f'(x) = ab/(b+x)^2$  – always positive  $\rightarrow$  monotonic, **no** critical point, saturates instead of peaking

- **Sticking point:** critical point ( $f' = 0$ )  $\neq$  root ( $f = 0$ ) – Ricker has both, at different  $x$  (root at  $x = 0$ , peak at  $x = 1/b$ )

#### 5. R convention + scale parameters (~5 min)

- Bolker's own rule: **x** first argument, params after, with defaults
- *this is why* `function(a, b, x)` in this morning's WOD 1a was buggy
- matters for `supply()`: vector element fills the *first* open slot
- **preview:** Lab 2 Act 2 asks you to write new bestiary functions “following the same pattern (**x** first, then parameters)”
- **scale parameter:**  $bx$  or  $x/c$  stretches/shrinks the  $x$ -axis; Lab 2's flexible Ricker reparameterizes  $b \rightarrow p$  (peak location directly) for exactly this reason

#### 6. Roadmap (~2 min)

- Wednesday: the bestiary itself (linear, power, exponential, hyperbolic, sigmoid) – same tools, now cataloging specific families
- **Reminder before they leave:** Lab 2 Act 2 needs today's limit trick for Holling IV

**If time: discussion seed**

- *Pick a function from your own field that saturates. What plays the role of “a” (the asymptote) – what caps it, mechanistically?*